Why a Two-Dimensional Ferrofluid Simulation Reproduces a Three-Dimensional Experiment

Technical note on the LBM validation against Bacri *et al.* (1995), Fig. 3

Summary. A planar (2D) lattice-Boltzmann simulation reproduces the theoretical curves of Bacri *et al.* to within 0.011 in g across 30 points. This note explains why no re-derivation of the magnetisation equation was needed; why the magnetic susceptibility (and with it the demagnetising field) matters as soon as the field stops being uniform; and exactly which future changes *would* force a re-derivation.

1. Why the three-dimensional equations are the correct ones

1.1 The word "dimension" is doing two different jobs

The confusion this note resolves comes from one word being used for two unrelated things.

The flow is two-dimensional. The velocity field lives in a plane, $\mathbf{v} = (v_x, v_y, 0)$, nothing varies along z, and the vorticity[1] points out of the plane.

The particle orientations are three-dimensional. Each magnetic nanoparticle carries a magnetic moment[2] whose direction is a point on the unit sphere. Brownian rotation[3] moves it over that sphere. Averaging the Boltzmann weight $e^{\xi \cos\theta}$ over the sphere is what produces the Langevin function[4] $L(\xi)$.

These are different objects at different scales. Flattening the first does nothing to the second, because a z-invariant flow does not reach down and forbid a nanoparticle from tumbling out of the plane.

A useful way to put it: the simulation is not a two-dimensional *universe*. It is a three-dimensional system with translational symmetry along z. We are not solving different equations — we are solving the same 3D equations restricted to a symmetry sector.

1.2 The governing equations

The magnetisation obeys the effective-field (Martsenyuk–Raikher–Shliomis) equation. Writing $\boldsymbol{\xi}_e$ for the effective field[5] and $L_e = L(\xi_e)$, $\xi_e = |\boldsymbol{\xi}_e|$, $\mathbf{n} = \boldsymbol{\xi}_e/\xi_e$:

$$\mathbf{M} = \varphi M_s \, L(\xi_e)\, \mathbf{n}, \tag{1}$$

$$\frac{d}{dt}(L_e \mathbf{n}) = \boldsymbol{\Omega} \times (L_e \mathbf{n}) - \frac{1}{\tau_B}\frac{L_e}{\xi_e}(\boldsymbol{\xi}_e - \boldsymbol{\xi}) - \frac{S_2(\xi_e)}{2\tau_B\, \xi_e^2}\, \boldsymbol{\xi}_e \times (\boldsymbol{\xi}_e \times \boldsymbol{\xi}), \tag{2}$$

with $S_2(\xi_e) = 1 - 3L_e/\xi_e$ and $\boldsymbol{\xi} = \mu_0 m \mathbf{H}/k_B T$ the true field.

[1] **Vorticity $\boldsymbol{\Omega} = \frac{1}{2}\nabla \times \mathbf{v}$**: a vector measuring how fast a small blob of fluid spins as it moves. In a planar flow it always points out of the plane.

[2] **Magnetic moment $\mathbf{m}$**: the strength and direction of a particle's own tiny magnet. Here it is frozen to the crystal, so the moment can only turn if the whole particle turns.

[3] **Brownian rotation**: random tumbling caused by collisions with liquid molecules. Its characteristic time is $\tau_B = 3\eta V/k_B T$.

[4] **Langevin function** $L(\xi) = \coth\xi - 1/\xi$: the average alignment of freely rotating moments in a field, where $\xi = \mu_0 m H/k_B T$ compares magnetic energy to thermal energy.

[5] **Effective field $\boldsymbol{\xi}_e$**: a bookkeeping vector. The real, lagging magnetisation is written as if it were an equilibrium magnetisation belonging to this fictitious field, which turns a hard statistical problem into a tractable ODE.

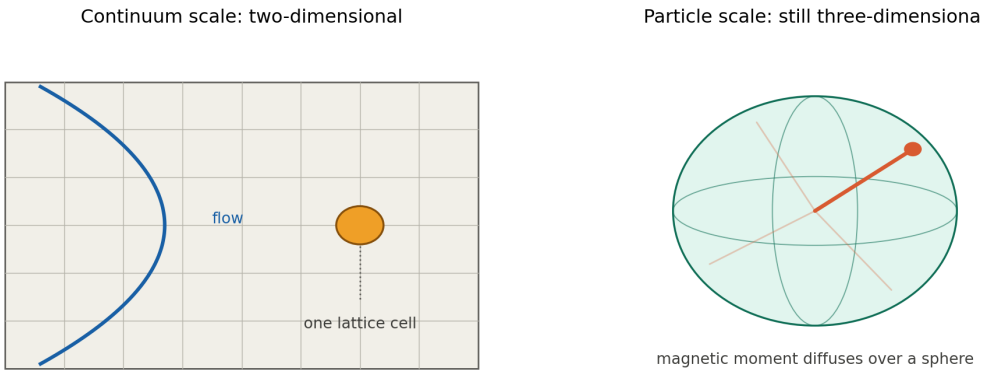


Figure 1: Left: the flow lives on a plane, so a lattice cell is a two-dimensional object. Right: inside that cell sit billions of particles whose moments still explore a full sphere. The orientational statistics is a sub-grid closure, not a property of the lattice.

In words: the first term says the flow's rotation carries the magnetisation around with it. The second says the magnetisation relaxes towards the field over the Brownian time τ_B. The third describes the fact that relaxation is slower across the field direction than along it.

1.3 The plane is an exact invariant manifold

Now the central claim. Take

$$\mathbf{H} = H\hat{x}, \qquad \mathbf{\Omega} = \Omega\hat{z}, \qquad \mathbf{M} \text{ in the } xy\text{-plane}.$$

Ask whether M_z can grow from zero. Check each term of (2):

$$\mathbf{\Omega} \times \mathbf{M} = \Omega\hat{z} \times (M_x\hat{x} + M_y\hat{y}) = \Omega(-M_y\hat{x} + M_x\hat{y}) \quad \Rightarrow \text{ in-plane}, \tag{3}$$

$$\mathbf{M} \times (\mathbf{M} \times \mathbf{H}) = \mathbf{M}(\mathbf{M}\cdot\mathbf{H}) - \mathbf{H}|\mathbf{M}|^2 = (-HM_y^2,\ HM_xM_y,\ 0) \quad \Rightarrow \text{ in-plane}, \tag{4}$$

$$\mathbf{M}_0 \propto \mathbf{H} = H\hat{x} \quad \Rightarrow \text{ in-plane}. \tag{5}$$

Every term has zero z-component, therefore

$$\boxed{\dot{M}_z \equiv 0 \quad \text{whenever} \quad M_z = 0} \tag{6}$$

In words: if the magnetisation starts in the plane, it stays in the plane forever. The plane is an *invariant manifold*[6] of the unmodified three-dimensional equation.

This is why there is nothing to re-derive. The 2D equations are not a new model; they are the 3D model evaluated on a subspace it already preserves. The reduction is exact, not an approximation, and it introduces no new functions and no new coefficients.

1.4 The reduced equations actually solved

Decomposing (2) into components parallel and perpendicular to $\mathbf{n}$ gives the form implemented in the code:

$$\frac{d\boldsymbol{\xi}_e}{dt} = \underbrace{\Omega\,(\hat{z} \times \boldsymbol{\xi}_e)}_{\text{carried by the flow}} - \underbrace{\frac{L_e}{L'(\xi_e)\,\tau_B}\left(1 - \frac{\mathbf{n}\cdot\boldsymbol{\xi}}{\xi_e}\right)\mathbf{n}}_{\text{relaxation along } \mathbf{n}} + \underbrace{\frac{1}{2\tau_B}\left(\frac{\xi_e}{L_e} - 1\right)[\boldsymbol{\xi} - \mathbf{n}(\mathbf{n}\cdot\boldsymbol{\xi})]}_{\text{relaxation across } \mathbf{n}} \tag{7}$$

[6] **Invariant manifold**: a surface that solutions never leave. If you start on it you stay on it, exactly — not approximately.

1.5 Verification: the reduction reproduces Bacri's equations exactly

Equation (2) is poorly reproduced in the published text, so rather than trusting a reading of it we verify (7) against the two equations Bacri actually integrate.

Step 1 — switch off the flow. Set $\Omega = 0$, so $\boldsymbol{\xi}_e = \xi_e \hat{x}$ and the transverse term vanishes:

$$\tau_B \frac{d\xi_e}{dt} = -\frac{L_e}{L'(\xi_e)}\left(1 - \frac{\xi_0 \cos \omega t}{\xi_e}\right) = -\left(\frac{d \ln L_e}{d\xi_e}\right)^{-1}\left(1 - \frac{\xi_0}{\xi_e}\cos \omega t\right)$$

which is Bacri Eq. (7), exactly.

Step 2 — switch the flow back on, weakly. Write $\boldsymbol{\xi}_e = \xi_e \hat{x} + \delta \hat{y}$ with $\delta = \xi_e F \tau_B \Omega$, and keep first order in $\Omega\tau_B$. The transverse field is $\boldsymbol{\xi}_\perp \simeq -(\delta \xi_t/\xi_e)\hat{y}$ with $\xi_t = \xi_0 \cos \omega t$, and the y-component of (7) gives

$$\dot{\delta} = \Omega \xi_e + \frac{\delta\, \dot{\xi}_e}{\xi_e} - \frac{\xi_t \delta}{2\tau_B \xi_e}\left(\frac{\xi_e}{L_e} - 1\right).$$

Substituting $\delta = \xi_e F \tau_B \Omega$, the terms in $\dot{\xi}_e$ cancel and, after dividing by $\Omega \xi_e$,

$$\tau_B \frac{dF}{dt} = 1 - \frac{1}{2}\left(\frac{1}{L_e} - \frac{1}{\xi_e}\right)\xi_0 F \cos \omega t$$

which is Bacri Eq. (8), exactly — including the grouping $(1/L_e - 1/\xi_e)$. This also settles the ambiguous factor in (2) as $S_2 = 1 - 3L_e/\xi_e$.

1.6 From magnetic torque to a change in viscosity

The magnetisation and the field are not parallel, so there is a torque[7] on the fluid, entering the momentum equation as

$$\rho \frac{d\mathbf{v}}{dt} = -\nabla p + \mu_0 (\mathbf{M}\cdot\nabla)\mathbf{H} + \eta \nabla^2 \mathbf{v} + \tfrac{1}{2}\mu_0 \nabla \times (\mathbf{M} \times \mathbf{H}). \tag{8}$$

Averaging over one field period, Bacri Eq. (9) gives $\langle \mathbf{M} \times \mathbf{H} \rangle = -6\eta\varphi g\, \boldsymbol{\Omega}$, and with $\Omega = -\tfrac{1}{2}\partial_y v_x$,

$$\langle F_x \rangle = -\tfrac{1}{2}\mu_0\, \partial_y \langle M_y H_x \rangle = -3\eta\varphi g\, \partial_y \Omega = \tfrac{3}{2}\, \eta\varphi g\, \frac{\partial^2 v_x}{\partial y^2}. \tag{9}$$

This is the step that makes the whole comparison possible. The averaged magnetic force contains *the same second derivative* as ordinary viscous friction, so the two merge:

$$0 = F_b + \eta\big(1 + \tfrac{3}{2}\varphi g\big)\, \frac{\partial^2 v_x}{\partial y^2} \qquad \Longrightarrow \qquad \Delta\eta = \tfrac{3}{2}\eta\varphi g.$$

The magnetic force does not add a new kind of physics to the flow — it simply **renames** η. That is why one number g captures everything, and why the velocity profile stays a clean parabola.

[7] **Torque $\mathbf{M} \times \mathbf{H}$**: twisting effort per unit volume. If $\mathbf{M}$ and $\mathbf{H}$ were parallel it would be zero and there would be no effect at all.

1.7 The one constant that had to be right

Everything above fixes the *shape* of the curves. The *height* comes from a single consistency requirement. Because $\tau_B = 3\eta V/k_BT$ and $m = M_sV$ refer to the same particle in the same liquid, the magnetisation scale is not free:

$$M_s = \frac{3\,\eta\,\varphi_p}{\mu_0\tau_B}\cdot\frac{\xi_0}{H_0}. \tag{10}$$

In words: the viscosity that resists a particle's rotation must be the same viscosity as the fluid being simulated. This is a physical constraint, not a tuning parameter. Using the wrong coefficient leaves every curve correctly shaped but uniformly too high — the failure mode most easily mistaken for success.

1.8 Why the pipe and the slab give the same answer

Bacri's experiment used a cylindrical capillary; the simulation uses a planar channel. Three reasons this does not matter.

(a) Fig. 3 is theory, not experiment. It is integrated from Bacri Eqs. (7)–(8). That theory is a *local, single-particle* description: it contains no pipe radius, no channel height, no wall. It is a constitutive law[8] and materials do not know what container they are in.

(b) The pipe is the planar problem, wrapped. In the capillary the field is axial, $\mathbf{H} = H\hat{x}$; the vorticity of pipe flow is azimuthal, $\boldsymbol{\Omega} = \Omega\hat{\theta}$; and the induced magnetisation follows $\boldsymbol{\Omega}\times\hat{h} \propto \hat{\theta}\times\hat{x} = \hat{r}$, i.e. radial. So at every point the magnetisation lives in the two-dimensional (x, r) plane with the vorticity perpendicular to it — identical in structure to the slab under $x\to x$, $r\to y$, $\theta\to z$.

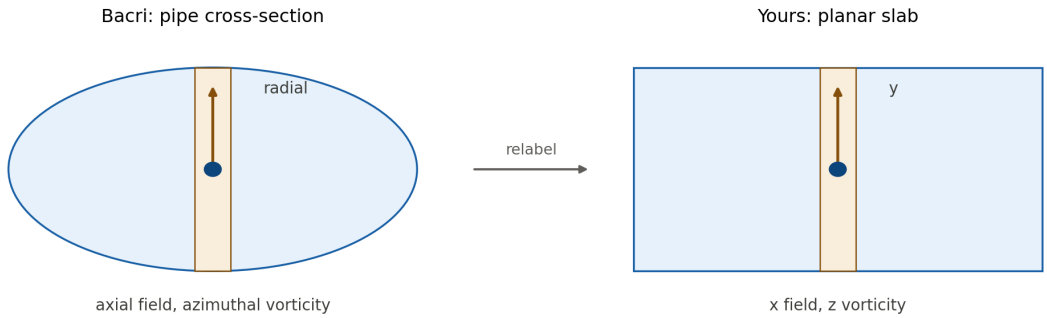


Figure 2: A fluid element in the pipe experiences the same local configuration as one in the slab. The shaded strip is the plane the magnetisation lives in; the dot marks the vorticity pointing out of it.

(c) The geometry cancels exactly from the measurement. The two geometries obey different flow laws ($Q \propto \Delta p R^4/\eta$ for a pipe, $Q = \rho S F/2\eta$ for a slab), but the reported quantity is a *ratio* taken in one and the same apparatus:

$$g = \frac{2}{3\varphi_p}\left(\frac{\eta}{\eta_0} - 1\right) = \frac{2}{3\varphi_p}\left(\frac{Q_0}{Q} - 1\right). \tag{11}$$

At fixed driving, $\eta/\eta_0 = Q_0/Q$. The R^4 cancels; the S, the ρ and the F cancel. The geometric factor is not approximately negligible — it is *exactly absent* from the observable.

[8] **Constitutive law**: a statement about the material itself (like "this liquid has viscosity η"), as opposed to a boundary-value problem, which is about the container. One does not re-derive the viscosity of water for a pipe versus a slab.

1.9 Result

Across 30 points spanning $\xi_0 \in [1, 20]$ and $\omega\tau_B \in [0.1, 2]$ the agreement is:

$\omega\tau_B$	0.1	0.5	1.0	1.5	2.0
$\max \lvert g_{\mathrm{LBM}} - g_{\mathrm{theory}} \rvert$	0.0103	0.0087	0.0036	0.0033	0.0005

The residual *shrinks* as frequency rises, which is the opposite of what a discretisation error would do. It is the signature of the $O\big((\Omega\tau_B)^2\big)$ terms that Bacri discard when reducing (2) to Eqs. (7)–(8) but which the simulation retains. In other words the small gap measures the theory's approximation, not the simulation's error.

2. Magnetic susceptibility and the demagnetising field

2.1 The definition and its size here

The susceptibility[9] is

$$\chi = \frac{M_x}{H_x} = \frac{M_s L(\xi_0)}{H_0} = 0.15\,\xi_0 L(\xi_0) \quad \text{for the parameters used here.} \tag{12}$$

ξ_0	1	5	20
χ	0.05	0.51	2.85

In words: at weak fields the fluid is almost invisible magnetically. At strong fields it magnetises nearly three times as hard as the field driving it — it has become a magnet in its own right, and its own field can no longer be ignored.

2.2 A magnetised fluid changes the field it sits in

The field must satisfy magnetostatics[10]

$$\nabla \times \mathbf{H} = 0, \qquad \nabla\cdot(\mathbf{H} + \mathbf{M}) = 0. \tag{13}$$

The flow-induced magnetisation is $M_y \propto \Omega(y)$, and for Poiseuille flow Ω is linear in y, so $\partial_y M_y \neq 0$. Solving (13) with $\partial_x \equiv 0$:

$$\partial_y(H_y + M_y) = 0 \;\Rightarrow\; H_y + M_y = \text{const} \xrightarrow[\text{about the centreline}]{\text{antisymmetry}} \boxed{H_y = -M_y} \tag{14}$$

In words: the sideways magnetisation piles up — magnetic north on one wall, south on the other — and the resulting poles create a field pointing backwards inside the fluid. This extra field is called the demagnetising field.[11]

[9] **Magnetic susceptibility** $\chi = M/H$: how strongly the material magnetises in response to a field. $\chi \ll 1$ means the fluid barely responds and barely alters the field; $\chi \sim 1$ means the fluid's own magnetisation is as large as the applied field.

[10] **Magnetostatics**: the rules a magnetic field obeys when nothing is changing fast. $\nabla\times\mathbf{H} = 0$ allows a potential $\mathbf{H} = -\nabla\psi$; $\nabla\cdot(\mathbf{H}+\mathbf{M}) = 0$ says magnetic field lines never begin or end.

[11] **Demagnetising field**: the field a magnetised body produces inside itself, always opposing its own magnetisation. It depends on the body's shape, which is why it is sometimes called a shape effect.

2.3 Why it cannot be ignored once the field is non-uniform

Substituting $H_y = -M_y$ into the torque:

$$(\mathbf{M} \times \mathbf{H})_z = M_x H_y - M_y H_x = -M_y(M_x + H_x) = -M_y H_x\,(1 + \chi). \tag{15}$$

In words: the measured effect is multiplied by $(1 + \chi)$. From the table above that factor runs from 1.05 at $\xi_0 = 1$ to 3.85 at $\xi_0 = 20$ — it grows along exactly the axis Fig. 3 sweeps. It is therefore not a constant offset that could be divided out; it *distorts the shape* of the curve.

A second consequence: a non-uniform field revives the Kelvin force[12] in (8), which is identically zero in a uniform field.

2.4 Limitations of the model used in the Bacri paper

1. **The demagnetising field is neglected.** Eqs. (3)–(6) of the paper use $\mathbf{H}$ as the *applied* field and never solve (13). Since the induced magnetisation is radial in their capillary and has a non-zero divergence, poles do form and $H_r = -M_r$ there too. The omission is an approximation, not a consequence of their geometry. At $\varphi = 0.2$ the neglected factor is $(1 + \chi) \sim 3$–4, and this is plausibly part of why their own experimental Fig. 2 does not match their theoretical Fig. 4 in detail.
2. **A dilute closure used at 20% concentration.** The coefficient $6\eta\varphi$ comes from Stokes drag on an *isolated* sphere. Shliomis notes that the rotational viscosity and Einstein's $\frac{5}{2}\eta\varphi$ are the same order, i.e. this is a dilute-suspension result. It is applied at $\varphi = 0.2$, where particles are close enough to interact hydrodynamically and magnetically.
3. **Rigid dipoles only.** The model assumes the moment can reorient only by turning the whole particle. Small particles also relax by Néel rotation[13], which produces no viscous effect at all. Valid for the paper's Co-ferrite; wrong for fine magnetite.
4. **One particle size.** A single τ_B and a single m. Real ferrofluids are polydisperse and $\tau_B \propto d^3$, so the true response is a broad superposition.
5. **Truncation at first order in $\Omega\tau_B$.** Eqs. (7)–(8) drop $O\big((\Omega\tau_B)^2\big)$. This is the source of the residual reported in §1.8.
6. **No spin diffusion.** Couple-stress terms that would smooth rapid spatial variation of $\omega_p - \Omega$ are omitted. Harmless in smooth shear flow; questionable near obstacles or interfaces.

2.5 What adding the field solver (h lattice) improves

Restoring a solver for (13) — in practice an extra lattice for the potential ψ, with $\mathbf{H} = \mathbf{H}_{\text{applied}} - \nabla\psi$ — removes limitation (1) and unlocks the physics that depends on it:

- **Correct torque at high field.** The factor $(1 + \chi)$ in (15) is computed rather than assumed away.
- **The Kelvin force becomes available.** Required for any magnetically pumped or magnetically steered flow, since $(\mathbf{M}\cdot\nabla)\mathbf{H}$ is zero without field gradients.

[12] **Kelvin force** $\mu_0(\mathbf{M}\cdot\nabla)\mathbf{H}$: the pull a magnetised fluid feels towards regions of stronger field — the same force that drags iron filings to a magnet. It vanishes identically in a uniform field.

[13] **Néel relaxation**: the moment flips inside a stationary particle by hopping over the crystal's anisotropy barrier. Such particles produce no torque on the liquid and so no rotational viscosity.

- **Interfaces behave correctly.** At a free surface $\mathbf{M}\cdot\hat{n} \neq 0$ creates surface poles. This *is* the mechanism of the normal-field (Rosensweig) instability — without the solver that physics is absent entirely, not merely inaccurate.
- **Obstacles and constrictions.** Any geometry that makes $\mathbf{M}$ non-uniform generates its own field, which then feeds back on the flow.

Two cautions when reconnecting it. First, the magnetisation update must read the *local* field; if it keeps using the analytic applied field while the force routine uses the solved field, the two are silently inconsistent. The applied part varies within a timestep and must stay analytic, while the demagnetising part may be held fixed across the step (a first-order-in-Δt approximation). Second, the permeability fed to the field solver must track the actual M/H, which varies by a factor of $\sim$50 over the sweep; a fixed value is badly wrong at low field.

Expect Fig. 3 agreement to break. Once the demagnetising field is on, the simulation should *disagree* with Fig. 3 by roughly $(1+\chi)$, because Bacri's theory omits it. That is correct behaviour, not a regression. Keeping a switch to disable the solver preserves the only closed-form validation available.

3. When the equations would have to be re-derived

3.1 The single decisive test

Everything in §1.3 rests on one condition:

$$\boxed{\mathbf{H} \text{ lies in the plane of the flow}} \tag{16}$$

Because the vorticity of any planar flow is automatically perpendicular to that plane, an in-plane field guarantees $\mathbf{H} \perp \boldsymbol{\Omega}$, and the invariant-manifold argument holds. Nothing else about the geometry enters.

3.2 Cases that DO require a full re-derivation

1. **A field with a component out of the plane.** If $H_z \neq 0$ then $\mathbf{M} \times (\mathbf{M} \times \mathbf{H})$ acquires a z-component, M_z becomes dynamically active, and (7) is no longer exact. The full three-component equation and a 3D solver are required.

   A concrete casualty: the classical result $\Delta\eta \propto \sin^2\beta$ (with β the angle between $\mathbf{H}$ and $\boldsymbol{\Omega}$) can never be tested in a planar solver, because $\beta = 90°$ identically. Rotating the field *within* the plane changes nothing. McTague's observation that $\Delta\eta_{\parallel} = 2\Delta\eta_{\perp}$, and the $\Delta\eta = 0$ case for $\mathbf{H} \parallel \boldsymbol{\Omega}$, are both out of reach.

2. **Resolving individual particles on the lattice.** At present the particles are sub-grid and the drag coefficient $6\eta\varphi$ is supplied from 3D Stokes flow past a sphere. If particles were resolved they would be infinite cylinders, whose drag differs, and the coefficient would have to be re-derived.

3. **Moments physically confined to a plane.** With a strong easy-plane anisotropy, or a pinned monolayer, the orientational space really is a circle and the equilibrium magnetisation becomes $I_1(\xi)/I_0(\xi)$ instead of $L(\xi)$ — changing the weak-field slope from $\xi/3$ to $\xi/2$. Note this is a *different material*, not a 2D reduction of the present one.

4. **Flow leaving the plane.** Any $v_z \neq 0$ or any z-dependence breaks the translational symmetry the whole construction rests on.

3.3 Cases that do NOT require re-derivation

Frequently mistaken for the above, but requiring only additional code:

Change	What is actually needed
Different boundary conditions (moving walls, inlet/outlet, obstacles)	New BC routines. The wall treatment currently hard-codes stationary no-slip, which also feeds the vorticity used by the magnetisation update.
Non-uniform or gradient applied field	The field solver, and the magnetisation update must read the local field.
Tilting the field within the plane	One line. Note that g is unchanged by an in-plane rotation, for the reason in §3.2.
Complex geometry: bends, constrictions, obstacles	Solid masks and curved boundaries; the field solver; a new way to measure the result.
Free surfaces and thin films	The phase field (already present) plus the field solver. Essential, since interface poles *are* the instability mechanism.
Strong shear, $\Omega\tau_B \sim 1$	Nothing. The implemented equation is the full (2), not the perturbative Eqs. (7)–(8), so it is valid at arbitrary $\Omega\tau_B$. Only the *reference curve* requires weak shear.

3.4 Limitations that remain regardless

These do not require re-derivation but should be stated when reporting results:

- **Demagnetising factors are those of an infinite slab**, $N_y = 1$, $N_x = N_z = 0$ — not those of a real rectangular channel.
- **Obstacles are infinite cylinders**, not spheres, so drag coefficients are cylinder values.
- **Surface patterns become stripes.** The Rosensweig instability forms a hexagonal lattice in the plane of a real film; a planar cross-section yields a periodic ridge array. The onset threshold and wavelength are right, the pattern symmetry is not.
- The physical omissions of §2.4 (items 2–6) are inherited from the model and are not fixed by any solver.

Bottom line. A planar solver with an in-plane field uses the three-dimensional magnetisation equation correctly and without modification, because the plane is an exact invariant manifold of that equation and the coefficients in it describe particles, not the lattice. Geometry cancels from the reported observable. What genuinely changes the physics is an out-of-plane field component; almost everything else is a matter of adding code, chiefly the magnetostatic field solver.